

Andy Rodriguez

Senior Data Engineer | Data Warehouse & Platform Specialist

Summary

Senior Data Engineer with 9+ years of expertise building **scalable ETL/ELT pipelines** and **cloud-native data platforms** for retail, healthcare, and advertising. Proven in developing **real-time streaming analytics** and **modern data warehouses** using **Snowflake, Redshift, dbt, Airflow, Spark, Kafka**, and **Python** to support high-impact **BI dashboards** and **ML feature workflows**. Delivered measurable results including significant **latency reduction**, **compute cost optimization**, and enhanced **data quality and governance** that empower data-driven decisions and enable self-service analytics.

Skills

Programming Languages: Python, Scala, Java, PHP, Shell Script

Databases: SQL, NoSQL, PostgreSQL, MongoDB, Oracle, Snowflake, Redshift, BigQuery, Iceberg, Delta Lake

Storage & Data Lakes: S3, ADLS, Parquet, ORC, Avro

Data Warehousing & Transformation: dbt, ELT Architecture, Dimensional Modeling, Analytics-Ready Datasets

Pipeline & Orchestration: Apache Airflow, Dagster, Fivetran, ETL/ELT Development, Batch & Streaming Pipelines, Workflow Optimization, Event-Driven Ingestion, Kubernetes

Cloud & Infrastructure: AWS (S3, Glue, Lambda, EMR, IAM, CloudWatch, SageMaker, CloudFormation), Terraform, CI/CD (GitHub Actions), Infrastructure as Code

Big Data & Distributed Processing: Apache Spark, PySpark, High-Volume Data Processing, Performance Optimization, Kafka

Data Governance & Observability: Unity Catalog, Prometheus, Grafana, Data Validation & Testing, A/B Testing

BI Collaboration & Analytics Enablement: Looker, Tableau, Power BI, Sigma, Metrics Standardization, Analytics Enablement, Business Stakeholder Support

Machine Learning & MLOps: Scikit-learn, TensorFlow, NumPy, Pandas, MLOps

Professional Experience

Walmart Connect | Bentonville, AR

Aug 2021 – Present

Senior Data Engineer

- Developed and maintained **high-volume ETL/ELT pipelines** in **Snowflake**, processing **billions of daily impressions, clicks, and transaction events**, powering **cross-channel attribution** and advertiser-facing dashboards, reducing **data latency from hours to under 15 minutes**.
- Designed **near-real-time streaming pipelines** with **Kafka** and **Spark/PySpark**, orchestrated via **Airflow**, to deliver **audience segmentation**, campaign performance metrics, and **real-time analytics** for Walmart DSP campaigns.

- Built **A/B testing and experimentation pipelines**, combining ad exposure, retail transactions, and behavioral signals, enabling **statistically valid incremental lift measurement** and **ROAS optimization**, increasing campaign effectiveness and optimization speed by **3x**.
- Implemented **data governance frameworks** using **Unity Catalog**, enforcing **fine-grained access control**, **PII anonymization**, and **GDPR compliance**, ensuring **enterprise-grade data security** across multiple data pipelines.
- Engineered **semantic-layer-ready datasets** and **self-service dashboards** in **Looker**, **Tableau**, and **Sigma**, enabling marketing, analytics, and product teams to explore KPIs independently, reducing ad-hoc data requests by **40%**.
- Partnered with **ML and data science teams** to productionize **feature pipelines** for **bid optimization** and audience targeting, leveraging **Python**, **Pandas**, **Scikit-learn**, and **AWS SageMaker**, enabling automated model retraining for real-time DSP scoring.
- Containerized **data pipelines** with **Kubernetes** and implemented **CI/CD deployment pipelines** using **Terraform** and **CloudFormation**, standardizing staging and production workflows across multiple high-traffic environments.
- Optimized **Snowflake warehouse performance** with **clustering**, **micro-partitioning**, and **workload isolation**, reducing compute costs by **25%** while maintaining high throughput for advertiser reporting during peak seasonal campaigns.
- Built **observability dashboards** with **Prometheus** and **Grafana**, implementing **SLA-based alerting**, reducing pipeline failure detection time by **70%** and improving overall reliability.
- Developed and maintained **event-driven ingestion frameworks** for Walmart DSP feeds, handling **high-volume bid, impression, and conversion logs** while implementing **partitioning and workload optimization** to improve throughput and reduce costs.
- Collaborated with **product managers, analytics, and business teams** to define data requirements, validate KPIs, and ensure pipelines and dashboards delivered **actionable insights**, aligning technical work with business goals.
- Mentored junior engineers on **pipeline design**, **Python scripting**, **SQL optimization**, and **modular architecture**, enhancing team capability for high-scale, multi-domain data projects.
- Initiated **modular pipeline architecture** improvements for reusable data processing components, enabling faster deployment of new campaigns and reducing development effort for repeated workflows by **30%**.
- Participated in **retail media platform modernization**, integrating **multi-source transactional and behavioral data**, designing efficient schema, and supporting **real-time advertising insights** across sponsored search, display, and CTV campaigns.

IBM Consulting | Armonk, NY

Dec 2017 – Jul 2021

Data Engineer

- **Architected** ETL/ELT pipelines in **Snowflake** and **Redshift**, handling **multi-client workloads** of billions of rows per day. Defined **partitioning and storage strategies** balancing **query performance**, **latency**, and **compute costs**, reducing overall pipeline runtime by **40%**.
- **Replatformed** batch and near-real-time workflows using **Airflow** and **Spark/PySpark**, evaluating **tradeoffs between micro-batching and streaming**, and implementing **backfill and failure recovery strategies** to maintain **99.9% SLA adherence**.
- **Introduced** dbt transformation models with **automated testing**, **schema evolution handling**, and **data contract governance**, standardizing **analytics-ready datasets** for multiple clients and improving dashboard accuracy by **35%**.

- **Drove cross-org alignment on ML feature pipelines**, collaborating with **data science teams** to define **feature granularity, normalization, and update frequency**, ensuring reproducibility and model accuracy for **churn prediction and fraud detection**.
- **Led platform-wide adoption of** cloud migration best practices from on-prem to **AWS**, including **Terraform- and CloudFormation-managed CI/CD pipelines**, influencing **multi-team roadmap and standards** for data platform modernization.
- **Established** observability frameworks using **Prometheus and Grafana**, defining **SLA metrics, alert thresholds, and dashboards** for multi-client pipelines, reducing failure detection time by **70%**.
- **Standardized** warehouse performance strategies across concurrent clients, including **clustering, micro-partitioning, and indexing**, enabling **multi-TB daily workloads** and **high-concurrency analytics**.
- **Institutionalized** reusable modular pipeline templates, accelerating deployment of new client projects and **reducing development effort by 30%**.
- **Led knowledge-sharing initiatives** and mentored junior engineers on **pipeline architecture, SQL optimization, and Python scripting**, improving team efficiency and enforcing **best practices across client pipelines**.
- **Partnered with product managers and client stakeholders** to define data requirements, validate KPIs, and ensure pipelines delivered **actionable insights** aligned with **business objectives and SLA commitments**.

Athenahealth | Watertown, MA

Jun 2015 – Nov 2017

Data Analyst

- Developed and automated **SQL extraction pipelines** and **Python transformation scripts** across clinical and operational datasets, reducing manual reporting by **50%** and enabling timely insights for **provider performance and care quality metrics**.
- Designed **analytics-ready datasets** by standardizing encounter, claims, and operational tables, improving **Tableau and Looker dashboard** accuracy and reducing data inconsistencies by **30%**.
- Built automated **data validation frameworks** that flagged anomalies in incoming feeds, enhancing **data quality** and ensuring **reliable KPI reporting** across operational and clinical dashboards.
- Created interactive **BI dashboards** integrating patient outcomes, appointment volumes, and claims metrics, enabling **actionable insights** and faster decision-making for cross-functional teams.
- Collaborated with **data engineering and product teams** to prototype reusable transformation modules and **modular ETL patterns**, improving workflow efficiency and preparing for future data engineering responsibilities.
- Mentored junior analysts on advanced **SQL querying** and **Python scripting**, improving team productivity and reinforcing **best practices** for reporting and automation.

Education

Bachelor of Science in Computer Science
University of Miami, Coral Gables, Florida
 Aug 2011 – May 2015